Src↓	Trg→	en	bn	hi	sw	yo
en	BLOOM	–	24.6	27.2	20.5	2.6
	M2M	–	23.0	28.1	26.9	2.2
bn	BLOOM	29.9	–	16.3	–	–
	M2M	22.9	–	21.8	–	–
hi	BLOOM	35.1	23.8	–	–	–
	M2M	27.9	21.8	–	–	–
sw	BLOOM	37.4	–	–	–	1.3
	M2M	30.4	–	–	–	1.3
yo	BLOOM	4.1	–	–	0.9	–
	M2M	4.2	–	–	1.9	–

(a) Low-resource languages

Src↓	Trg→	ca	es	fr	gl	it	pt
ca	BLOOM	–	28.9	33.8	19.2	19.8	33.0
	M2M	–	25.2	35.1	33.4	25.5	35.2
es	BLOOM	31.2	–	24.8	23.3	16.5	29.1
	M2M	23.1	–	29.3	27.5	23.9	28.1
fr	BLOOM	37.2	27.5	–	24.9	24.0	38.9
	M2M	28.7	25.6	–	32.8	28.6	37.8
gl	BLOOM	37.5	27.1	33.8	–	18.3	32.2
	M2M	30.1	27.6	37.1	–	26.9	34.8
it	BLOOM	31.0	25.4	31.4	20.2	–	29.2
	M2M	25.2	29.2	34.4	29.2	–	31.5
pt	BLOOM	39.6	28.1	40.3	27.1	20.1	–
	M2M	30.7	26.9	40.2	33.8	28.1	–

(b) Romance languages

Src ↓	Trg →	ar	en	es	fr	zh
ar	BLOOM	–	40.3	23.3	33.1	17.7
	M2M	–	25.5	16.7	25.7	13.1
	AlexaTM	–	41.8	23.2	35.5	–
en	BLOOM	28.2	–	29.4	45.0	26.7
	M2M	17.9	–	25.6	42.0	19.3
	AlexaTM	32.0	–	31.0	50.7	–
es	BLOOM	18.8	32.7	–	24.8	20.9
	M2M	12.1	25.1	–	29.3	14.9
	AlexaTM	20.8	34.6	–	33.4	–
fr	BLOOM	23.4	45.6	27.5	–	23.2
	M2M	15.4	37.2	25.6	–	17.6
	AlexaTM	24.7	47.1	26.3	–	–
zh	BLOOM	15.0	30.5	20.5	26.0	–
	M2M	11.55	20.9	16.9	24.3	–
	AlexaTM	–	–	–	–	–

(c) High-resource language pairs.

Src ↓	Trg →	en	fr	hi	id	vi
en	BLOOM	–	45.0	27.2	39.0	28.5
	M2M	–	42.0	28.1	37.3	35.1
fr	BLOOM	45.6	–	18.5	31.4	32.8
	M2M	37.2	–	22.9	29.1	30.3
hi	BLOOM	35.1	27.6	–	–	–
	M2M	27.9	25.9	–	–	–
id	BLOOM	43.2	30.4	–	–	–
	M2M	33.7	30.8	–	–	–
vi	BLOOM	38.7	26.8	–	–	–
	M2M	29.5	25.8	–	–	–

(d) High→mid-resource language pairs.

Table 8: 1-shot MT results (spBLEU) on the Flores-101 devtest set.

4.3.3 Flores

In the 1-shot setting, we test several language directions in the Flores-101 (Goyal et al., 2022) devtest set using the "xglm-source+target" prompt (Lin et al., 2021). The 1-shot example is randomly taken from the dev set. We separate out results for low-resource language pairs (Table 8a), between related languages of the Romance language family (Table 8b), high-resource language pairs (Table 8c) and high-to-mid-resource language pairs (Table 8d).

33. The truncation rule is specific to the "xglm-source+target" prompt and the fact that overgeneration consists of repeating the prompt pattern. Anything after a first newline or the regular expression pattern `= .+?:` is discarded.

Languages are classified as low-, mid- and high-resource depending on their representation in ROOTS. We compare to supervised results from the M2M-100 model (Fan et al., 2021) with 615M parameters, for which scores are computed by Goyal et al. (2022). Additionally, we compare to 32-shot AlexaTM results for high-resource language pairs (Soltan et al., 2022). Results are good across the board for both translation between high-resource languages and from high- to mid-resource languages, suggesting BLOOM's good multilingual capacity, even across scripts (here between Latin (or extended Latin), Chinese, Arabic and Devanagari scripts). Compared to the supervised M2M-100 model, results are often comparable and sometimes better in this 1-shot setting, and results are comparable in many cases to those of AlexaTM (even though AlexTM results are for 32-shot).

The translation quality for many of the low-resource languages is good, comparable to or even slightly better than the supervised M2M model. However, results are very poor between Swahili and Yoruba, languages that are present but under-represented in BLOOM's training data (<50k tokens each). This contrasts with the results for translation between Romance (and therefore related) languages, where results are good across-the-board, including for translation from Galician (glg), a language not included in the training data, but which shares many similarities with the other Romance languages, in particular with Portuguese (por). This however does question BLOOM's quality on those under-represented low-resource languages included in training.

4.4 Summarization

Figure 9 shows one-shot results for BLOOM models alongside OPT-175B for comparison. Each point represents a per-prompt score. The key takeaways are that BLOOM attains higher performance on multilingual summarization than OPT and that performance increases as the parameter count of the model increases. We suspect this is due to BLOOM's multilingual-focused training.

As discussed in Section 4.1, we report ROUGE-2 scores for the sake of comparability with prior work, and because there is a lack of alternatives for generation evaluation. However, we qualitatively observe that in many cases, the ROUGE-2 score understates the quality of the summaries generated by the systems.

4.5 Code Generation

The BLOOM pretraining corpus, ROOTS, consists of around 11% of code. In Table 9, we report benchmarking results of BLOOM on HumanEval (Chen et al., 2021). We find the performance of pretrained BLOOM models to be similar to that of the similar-sized GPT models trained on the Pile (Gao et al., 2020). The Pile contains English data and around 13% of code (GitHub + StackExchange), which is similar to the code data sources and proportions in ROOTS. The Codex models, which have solely been finetuned on code, are significantly stronger than other models. Multitask finetuned BLOOMZ models do not improve significantly over BLOOM models. We hypothesize this is due to the finetuning dataset, xP3, not containing significant amounts of pure code completion. Rather, xP3 contains code-related tasks, such as estimating the time complexity of a given Python code snippet. Additional analysis is provided in Muennighoff et al. (2022b).

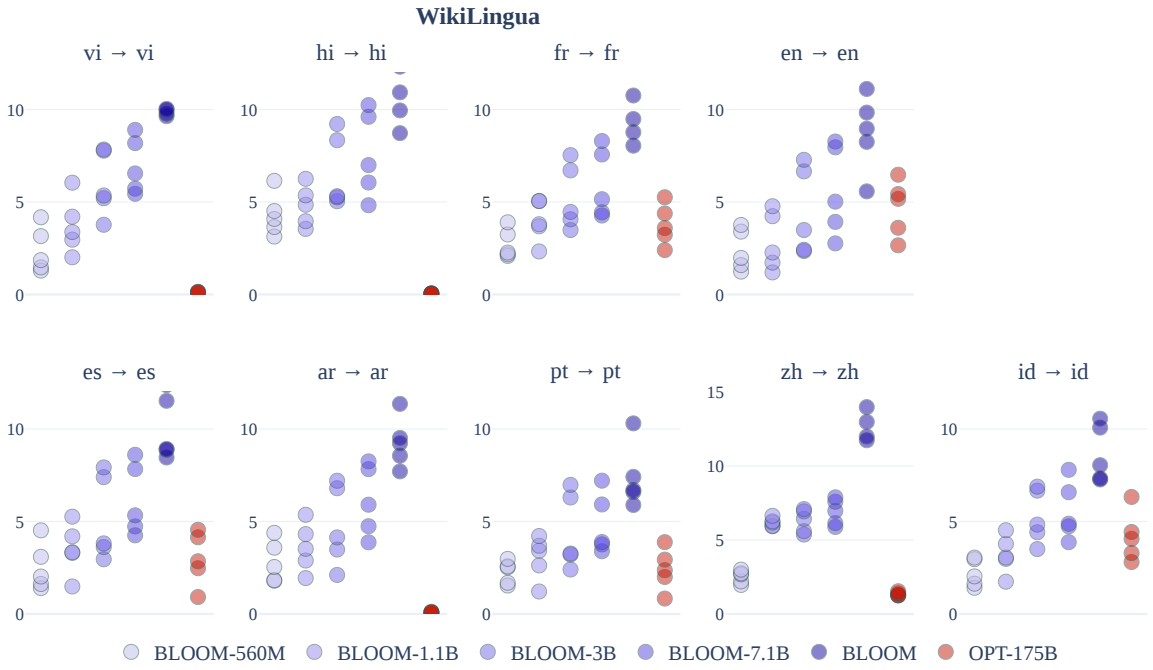

Figure 9: WikiLingua One-shot Results. Each plot represents a different language with per-prompt ROUGE-2 F-measure scores.

4.6 HELM benchmark

For completeness, we reproduce here evaluations from the HELM benchmark (Liang et al., 2022), which ran 5-shot evaluations of a variety of language models on English-only tasks. Despite the multilingual training, BLOOM is roughly on par in accuracy with previous-generation English-only models, such as GPT3-davinci v1 and J1-Grande v1, but behind more recent monolingual models such as InstructGPT davinci v2, Turing NLG v2, Anthropic-LM v4-s3, or OPT. Like other large language models of this size, it is not very well calibrated, but quite robust. Finally, on this benchmark, it is one of the best models for fairness, slightly more toxic than average in English, and average for bias.

4.7 Multitask Finetuning

Building on recent work on multitask finetuning (Sanh et al., 2022; Wei et al., 2021; Wang et al., 2022a) we explore using *multilingual* multitask finetuning to improve the zero-shot performance of the BLOOM model. We conducted multilingual multitask finetuning of BLOOM models using the xP3 corpus outlined in Section 3.1.4. We find that zero-shot performance significantly increases. In Figure 11, we compare the zero-shot performance of pretrained BLOOM and XGLM models with multitask finetuned BLOOMZ, T0 and mTk-Instruct (Wang et al., 2022b). BLOOM and XGLM performances are near the random baselines of 33% for NLI (XNLI) and 50% for coreference resolution (XWinograd) and